Part I: Choose the correct answer from the given alternatives and put your answer inside the shaded region only! (12 %).

C

1. Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:
   A) the shear
   B) the fractional change in volume
   C) the fractional change in length.
   D) the pressure
   E) the spring constant

B

2. An object hangs from a spring balance. The balance indicates 30N in air and 20N when the object is submerged in water. What does the balance indicate when the object is submersed in a liquid with a density that is half that of water?
   A) 20N B) 25N C) 30N D) 35N E) 40N

A

3. During the time that latent heat is involved in a change of state:
   A) the temperature does not change
   B) the substance always expands
   C) a chemical reaction takes place
   D) molecular activity remains constant
   E) kinetic energy changes into potential energy

E

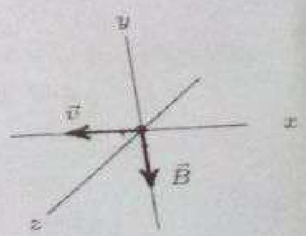
4. Temperature of a gas is due to:-
   A) Its heating value
   B) Surface tension of molecules
   C) Repulsion of molecules
   D) Attraction of molecules
   E) Kinetic energy of molecules

A D

5. An isolated charged point particle produces an electric field with magnitude E at a point 2 m away from the charge. A point at which the field magnitude is E/4 is:
   A) 1 m away from the particle
   B) 0.5 m away from the particle
   C) 2 m away from the particle
   D) 4 m away from the particle
   E) 8 m away from the particle

E

6. An electron moves in the negative x direction, through a uniform magnetic field in the negative y direction. The magnetic force on the electron is:
   A) in the negative x direction
   B) in the positive y direction
   C) in the negative y direction
   D) in the positive z direction
   E) in the negative z direction



D 7. When a forward bias is applied to a p-n junction the concentration of electrons on the p side:

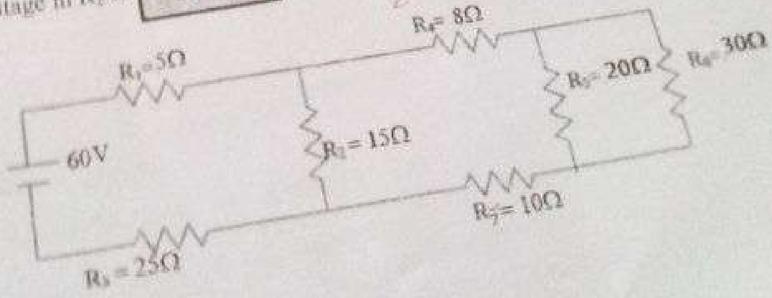
A) increases slightly
B) increases dramatically
C) decreases slightly
D) decreases dramatically
E) does not change

C 8. For a transistor to function as an amplifier,

A) both the EB and CB junctions must be forward-biased.
B) both the EB and CB junctions must be reverse-biased.
C) the EB junction must be forward- biased and the CB junction must be reverse-biased.
D) the CB junction must be forward- biased and the EB junction must be reverse-biased
E) none of the above

Part II: <u>Fill in the blank spaces with appropriate answers</u> (9 %)

1. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 2 cm. This pressure is transmitted by a liquid to a piston that has a radius of 4 cm and $1.4 \times 10^4 N$ is used to lift a car weight. The force that must be exerted is [illegible] N
2. A thermodynamic process which does not involve an exchange of heat between the system and the surrounding is adiabatic process
3. A 400 g object absorbing 2.5 kJ of heat has raised its temperature by 10^0C. The specific heat capacity of the object is 625 J/kg°C
4. A proton is moving in a circular orbit of radius 14 cm in a uniform magnetic field of magnitude 0.35 T directed perpendicular to the velocity of the proton. The orbital speed of the proton is 2.3 × 10 m/s
5. Rectifier is an electric circuit that converts AC to DC.
6. For the circuit shown in the Figure below, the equivalent resistance of the circuit is 40Ω and the voltage in R_2 is 15V



Part III. Give short and brief answers for the following questions. (9 %)

1. (a) What are elastic materials? (1.5 pt.)

- Elastic materials are materials that tends to get back to their original shape after an external force is removed.

(b) State Archimedes' principle (1.5 pt.)

→ ~~an object submerged in a water~~ the volume of an object that is submerged in a liquid is equal to the volume of displaced liquid.

→ < there is no way that two objects occupy the same space at the same time. >

→ ~~In otherwords the weight of~~ $F_b = S_l V_{dl} \vec{g}$

↳ in otherwords force of buoyancy is the difference between the weight of an object on air and the weight of an object when it is submersed in liquid.

2. Explain the similarities and differences between heat and work. (3 pt.)

Heat - the spontaneous ~~transfer~~ form of energy.
- is positive when absorbed and -ve when realeased
- is zero in adiabatic process
- is equal to workdone in cyclic process

Work - the nonspontaneous transfer of energy
- is positive when it is on the system and by the surrounding
and negative when it is by the system and on the surrounding.
- is zero in isochoric process.

3. State the following laws of electromagnetic induction:

(a) Faradays law (1.5 pt.)

→ states than an induced electromotive force is proportional to number of magnetic flux.

Induced emf $= -N \frac{\Delta \phi}{\Delta t}$

↳ faradays law

b) Lenz's law (1.5 pt.)

→ an induced emf is ~~equal in~~ opposite in direction from the magnetic flux

Induced emf $= -N \frac{\Delta \phi}{\Delta t}$

↳ the -ve sign, Lenz's law

Part IV. Solve the following problems by showing all the necessary steps. (20%)

1. Water circulates throughout a house in a hot water heating system. If the water is pumped at a speed of $5\ m/s$ through a $0.4\ m$ diameter pipe in the basement under a pressure of $3\times10^5\ Pa$, what will be the velocity and pressure in a $0.2\ m$ diameter pipe on the second floor $10\ m$ above? (4 pts.)

Given

$V_1 = 5\ m/s$

$d_1 = 0.4\ m$

$h_1 = 0$ basement

$P_1 = 3\times10^5\ Pa$

$d_2 = 0.2\ m$

$h_2 = 10\ m$

Required

$V_2 = ?$

$P_2 = ?$

$AV = \text{constant}$

$A_1V_1 = A_2V_2$

$\frac{\pi d_1^2}{4} V_1 = \frac{\pi d_2^2}{4} V_2$

$V_2 = \frac{d_1^2 V_1}{d_2^2} = \frac{(0.4)^2}{(0.2)^2}(5\ m/s)$

$= \left(\frac{0.16}{0.04}\times 5\right) m/s$

$= 20\ m/s$

Solution

$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$

$P_1 + \frac{1}{2}\rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g h_2$

$3\times10^5\ Pa + \frac{1}{2}(1000\ kg/m^3)(5\ m/s)^2 + (1000\ kg/m^3)(10\ m/s^2)(0) = P_2 + \frac{1}{2}(1000\ kg/m^3) V_2^2 + (1000\ kg/m^3)(10)(10)$

$3\times10^5\ Pa + 1.25\times10^4\ Pa = P_2 + 2\times10^5\ Pa + 10^5\ Pa$

$-P_2 = -3.125\times10^5\ Pa + 3\times10^5\ Pa$

$-P_2 = -0.125\times10^5\ Pa$

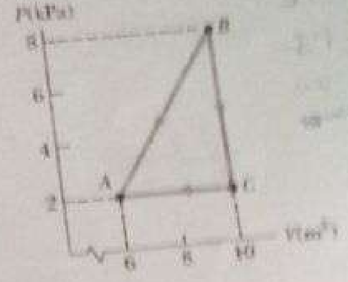
$P_2 = 1.25\times10^4\ Pa$

$V_2 = 20\ m/s$

$P_2 = 1.25\times10^4\ Pa$

4

2. A gas is taken through the cyclic process described as in the Figure below. (4 pts)
(a) Find the work done for the processes A to B, B to C, C to A and for the complete cycle
(b) Find the net energy transferred to the system by heat during one complete cycle



(a) cyclic process
$\Delta U = 0$
~~$W = -P\Delta V$~~

$W_{AB} = -(P_B - P_A)(V_B - V_A)$
$= -(8kPa - 2kPa)(10m^3 - 6m^3)$
$= -(6000Pa)(4m^3)$
$= -24000J$
$= -24kJ$

$W_{BC} = -P\Delta V$
$= -(P_B - P_C)(V_C - V_B)$
$= -(2kPa - 8kPa)(10m^3 - 10m^3)$
$= 0J$

$W_{CA} = -P\Delta V$
$= -2kPa(6m^3 - 10m^3)$
$= -2kPa(-4m^3)$
$= +8kJ = 8kJ$

$W_T = \frac{1}{2}$ area of the triangle
$= \frac{1}{2}(6kPa)(4m^3)$
$= \frac{1}{2}(24J)$
$= 12J$

1.5

25

Part I: Choose the correct answer from the given alternatives (15 pts). ***put your Answer in the shaded region only.!!!!***

C

1. A magnetic field exerts a force on a charged particle:
   A. always
   B. never
   C. if the particle is moving across the field lines
   D. if the particle is moving along the field lines
   E. if the particle is at rest

C

2. It is more difficult to measure the coefficient of volume expansion of a liquid than that of a solid because:
   A. no relation exists between linear and volume expansion coefficients
   B. a liquid tends to evaporate
   C. a liquid expands too much when heated
   D. a liquid expands too little when heated
   E. the containing vessel also expands

A

3. Heat is:
   A. energy transferred by virtue of a temperature difference
   B. energy transferred by macroscopic work
   C. energy content of an object
   D. a temperature difference
   E. a property objects have by virtue of their temperatures

E

4. The heat capacity of an object is:
   A. the change in its temperature caused by adding 1 J of heat
   B. the amount of heat energy that changes its state without changing its temperature
   C. the amount of heat energy per kilogram that raises its temperature by $1^{\circ}C$
   D. the ratio of its specific heat to that of water
   E. the amount of heat energy that raises its temperature by $1^{\circ}C$

A

5. During the time that latent heat is involved in a change of state:
   A. the temperature does not change
   B. the substance always expands
   C. a chemical reaction takes place
   D. molecular activity remains constant
   E. kinetic energy changes into potential energy

D

6. In an adiabatic process:
   A. the energy absorbed as heat equals the work done by the system on its environment
   B. the energy absorbed as heat equals the work done by the environment on the system
   C. the energy absorbed as heat equals the change in internal energy
   D. the work done by the environment on the system equals the change in internal energy
   E. the work done by the system on its environment is equal to the change in internal energy

B

7. The coefficient of linear expansion of iron is 1.0×10^{-5} per $^{\circ}C$. By what amount the surface area of an iron cube, with an edge length of 5.0 cm, will increase if it is heated from $10^{\circ}C$ to $60^{\circ}C$?
   A. $0.0125\ cm^2$ B. $0.025\ cm^2$ C. $0.075\ cm^2$ D. $0.15\ cm^2$ E. $0.30\ cm^2$

B

8. Faraday's law states that an induced emf is proportional to:
   A. the rate of change of the magnetic field
   B. the rate of change of the magnetic flux
   C. the rate of change of the electric field
   D. the rate of change of the electric flux
   E. none

D

9. What type of semiconductor results by doping silicon with trivalent atoms?
   A. N-type B. Intrinsic C. Superconductor D. P-type E. None

$\beta = 2\times10^{-5}$

$A_0 = (5\times10^{-2})^2 = 25\times10^{-4}$

$\Delta A = \beta A_0 \Delta T$
$= 2\times10^{-5} \times 25\times10^{-4} \times 50^{\circ}c$
$= 2.5\times10^{-6}\ m^2$

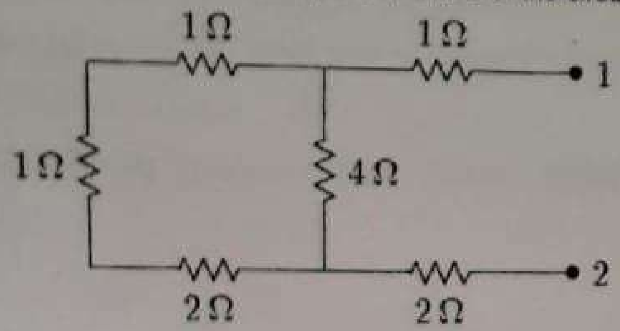
$2.5\times10^{-2}\ cm^2$

0.025

$N \frac{\Delta \phi}{\Delta t}$

$\frac{1}{4}+\frac{1}{4} = \frac{2}{4}$

10. The equivalent resistance between points 1 and 2 of the circuit shown is:



A. 3 Ω B. 4 Ω C. 5 Ω D. 6 Ω E. 7 Ω

Part II: Fill in the blank spaces with appropriate answers (10pts.) *put your Answer in the shaded region only.!!!!*

1. In Isobaric process the pressure remains constant as other variables are changing.
2. The temperature of a solid bar rises by 10^0C when it absorbs 1.20 kJ of energy by heat. The mass of the bar is 500 g. The specific heat of the bar is 240 J/kg°c
3. An object oscillates with simple harmonic motion along the x- axis. Its position varies with time according to the equation $x = (5.0\ \text{m})\cos\left(\frac{\pi}{2}t - \frac{\pi}{4}\right)$. Where t is in seconds and the angles in the parentheses are in radians. The period of the motion is 4 sec
4. Transistor is a semiconductor device used as a switch and an amplifier.
5. A converging lens with 50cm focal length forms a real image that is 2.5 times larger than the object. The object distance from the image is 3.5m

$Q = mc\Delta T$

$c = \frac{Q}{m\Delta T} = \frac{1.2\times10^3 J}{0.5kg\ 10°c} = 240$

$\omega = \frac{2\pi}{T}$ $\omega = \frac{\pi}{2}$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{\pi}{2}} = 4$

$M = \frac{i}{o}$

$\frac{i}{o} = 2.5$

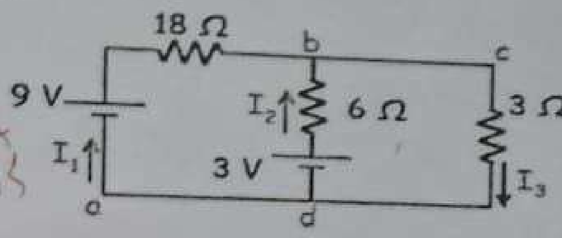
Part IV. Solve the following problems by showing all the necessary steps (20 pts)

1 For the circuit shown below, find

a) The currents in the three branches.

b) The potential difference between points b and d (5pts)

c) The power dissipated across the 3 Ω resistor.



a) At Junction b,

$$I_1 + I_2 = I_3 \quad \dots (1)$$

From loop a b d a,

$$9V - 18I_1 + 6I_2 - 3V = 0$$
$$6 - 18I_1 + 6I_2 = 0$$
$$18I_1 - 6I_2 = 6$$
$$3I_1 - I_2 = 1 \quad \dots (2)$$

From loop b c d b,

$$-3I_3 + 3V - 6I_2 = 0$$
$$2I_2 + I_3 = 1 \quad \dots (3)$$

$$\begin{cases} I_1 + I_2 = I_3 \\ 3I_1 - I_2 = 1 \\ 2I_2 + I_3 = 1 \end{cases}$$

Substituting eq (1) + eq. (3) gives

$$2I_2 + (I_1 + I_2) = 1$$
$$I_1 + 3I_2 = 1 \quad \dots (4)$$

Solving eq (2) and eq (4) Simultaneously,

$$3\times \begin{cases} 3I_1 - I_2 = 1 \\ I_1 + 3I_2 = 1 \end{cases}$$

$$+ \quad 10I_1 = 4$$
$$I_1 = 0.4\ A$$

using this value in eq (2) gives

$$I_2 = 3I_1 - 1$$
$$= 3(0.4) - 1$$
$$= 0.2A$$

using eq (1), $I_3 = I_1 + I_2 = 0.6\ A$

b) $\Delta V_{bd} = V_d - V_b = ?$

$$= 6I_2 - 3V$$
$$= 6(0.2) - 3V$$
$$= 1.2V - 3V$$
$$= -1.8V$$

∴ "d" is at lower potential than "b".

c) $P_{3\Omega} = I_3^2 R$

$$= (0.6A)^2(3\Omega)$$
$$= 1.08\ W$$

3. A train is moving parallel to a highway with a constant speed of 20.0 m/s. A car is traveling in the same direction as the train with a speed of 40.0 m/s. The car horn sounds at a frequency of 510 Hz, and the train whistle sounds at a frequency of 320 Hz.

(4 pts.)

a) When the car is behind the train, what frequency does an occupant of the car observe for the train whistle?

b) After the car passes and is in front of the train, what frequency does a train passenger observe for the car horn?

$V_t = 20 m/s$

$V_c = 40 m/s$

$f_c = 510 Hz$

$f_t = 320 Hz$

From Doppler's frequency shift

$$f' = f\left(\frac{v \pm v_o}{v \mp v_s}\right)$$

where v - is speed of sound in air
V_o - speed of observer
V_s - speed of source.

ⓐ: $f' = 320 \left(\frac{343 + 40}{343 + 20}\right) = 338 Hz$

b. $f' = 510 \left(\frac{343 + 20}{343 + 40}\right) = 483 Hz$